

Tanishq Rana

Full-Stack & Backend Software Engineer | Data & AI/ML Systems | Ex-Bain & ZS

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PROFESSIONAL SUMMARY

Software engineer and data professional with 2+ years of experience at Bain & Company and ZS Associates delivering scalable backend systems, full-stack applications, data pipelines, and AI-powered solutions. Skilled in Python, Golang, React, SQL, and AWS, with hands-on experience in microservices, ETL/ELT, Power BI, Tableau, and agentic AI systems. Master of Computer Science, University of Sydney. Shipped 5 live products across AI, full-stack, and developer tooling. Open to Software Engineering, Data Engineering, and AI/ML Engineering roles in Sydney.

CORE SKILLS

- **Languages:** Python, Golang, JavaScript, TypeScript, SQL, C#
- **Frontend & Full-Stack:** React.js, Next.js, Node.js, HTML/CSS, Tailwind CSS
- **Backend & APIs:** REST APIs, GraphQL, FastAPI, Express.js, Microservices, gRPC
- **AI / ML:** Machine Learning, PyTorch, LangGraph, AI Agent Development, Generative AI, Prompt Engineering, RAG, Hugging Face, MCP
- **Data Analysis & BI:** Power BI, Tableau, Advanced Excel, ETL/ELT Pipelines, Data Modelling, KPI Frameworks, Large-Scale Dataset Profiling
- **Databases:** PostgreSQL, SQL Server, Neo4j, Prisma, Convex, Databricks
- **Cloud & DevOps:** AWS, Azure, Docker, CI/CD, GitHub Actions, Linux, Git

PROFESSIONAL EXPERIENCE

ZS Associates

Sep 2022 – Jan 2024

Data & Software Engineer

New Delhi, India

- Engineered scalable data processing services in Golang and Python on AWS and Azure; built Power BI and Tableau dashboards and performed large-scale SQL analysis using Databricks to deliver insights across 5–10 enterprise client environments.
- Designed distributed microservices architecture defining service boundaries, data flows, and API contracts for a global reporting platform; contributed to ML pipeline workflows and training data preparation.
- Pioneered GenAI tooling (PyTorch, Copilot, Prompt Engineering) to automate reporting workflows and accelerate development velocity across the team.

Bain & Company

Feb 2022 – Aug 2022

Software Engineer

New Delhi, India

- Shipped RESTful APIs in C# and Node.js integrated into client-facing live production systems within one of the world's most demanding consulting environments.
- Delivered full software lifecycle — requirements, development, Docker containerisation, CI/CD on AWS, and production deployment — collaborating with React front-end teams under tight deadlines.

PROJECTS

SensAI – AI Career Coach

sens-ai-two-taupe.vercel.app

Next.js, React, PostgreSQL, Gemini AI, Clerk Auth — Full-stack AI coaching platform with ATS resume generation, AI mock interviews (1,000+ questions), and real-time industry insights across 50+ industries.

TeachSmart – AI Curriculum Platform

teachsmart-production.up.railway.app

React, Node.js, TypeScript, Groq AI — AI-powered resource finder for Australian teachers (Years 7–12); generates lesson plans, slide decks, and semester overviews with multilingual UI (8 languages).

AI PR Reviewer

github.com/Tran0408/AI-PR-Reviewer

Next.js, TypeScript, FastAPI, Python, PostgreSQL, Redis (ARQ), Docker, GitHub Actions — Full-stack app that automatically reviews GitHub PRs using an LLM; verifies HMAC webhooks, queues async jobs via Redis, posts Pydantic-structured inline review comments, and displays results in a Next.js dashboard. Deployed on Vercel and Railway with CI/CD.

Neo4j Agentic AI Prototype

LangGraph, Neo4j, FastAPI, Python, MCP — Agentic AI system converting natural language questions into validated Cypher queries with visual insights; modular FastAPI backend with schema-aware validation and MCP integration.

Splitr – Smart Expense Splitter

splitr-ten.vercel.app

React, Next.js, Convex, TypeScript — Real-time expense splitting app with smart settlement algorithm minimising payment paths, optimistic UI, automated reminders, and monthly spending digest.

EDUCATION

University of Sydney

Master of Computer Science

Completed Dec 2025
Sydney, NSW, Australia

Guru Gobind Singh Indraprastha University

Bachelor of Technology – Computer Science

2018–2022
New Delhi, India